

USUGLI, Russian Federation

WMO: 307640

Lat: 52.654N

Lon: 115.259E

Elev: 646

StdP: 93.80

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.7	(c) -39.4	(d) -43.9	(e) 0.1	(f) -41.4	(g) -41.5	(h) 0.1	(i) -39.0	(j) 6.8	(k) -14.5	(l) 5.9	(m) -15.8	(n) 0.0	(o) 180	(p) 0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	29.2	18.6	27.3	17.9	25.5	17.2	20.4	26.3	19.3	24.8	18.3	23.4	2.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	14.5	22.9	17.5	13.5	21.5	16.4	12.6	20.7	62.1	26.5	58.2	25.0	54.9	23.6	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.5	6.3	5.3	-44.1	32.6	2.8	2.1	-46.1	34.1	-47.7	35.4	-49.3	36.6	-51.3	38.1
			-43.7	22.8	2.7	1.6	-45.7	24.0	-47.3	24.9	-48.8	25.8	-50.8	27.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.1	-27.2	-21.4	-11.3	0.2	8.0	15.0	17.9	15.4	8.1	-1.6	-15.6	-25.7
	DBStd	16.67	5.56	5.63	6.72	4.55	3.86	3.51	2.55	3.26	4.04	5.06	7.22	6.29
	HDD10.0	5394	1154	879	659	294	85	4	0	3	82	359	767	1107
	HDD18.3	7859	1412	1113	918	543	319	109	37	100	308	617	1017	1366
	CDD10.0	619	0	0	0	1	24	154	246	170	25	0	0	0
	CDD18.3	41	0	0	0	0	0	8	24	8	0	0	0	0
	CDH23.3	938	0	0	0	1	28	310	426	159	14	0	0	0
	CDH26.7	231	0	0	0	0	2	85	111	32	1	0	0	0
	WSAvg	1.5	0.6	0.8	1.6	2.6	2.5	1.8	1.5	1.4	1.7	1.6	1.0	0.5
Precipitation	PrecAvg	358	3	3	3	10	29	70	90	85	44	9	7	5
	PrecMax	474	12	9	9	30	64	145	196	186	102	35	25	16
	PrecMin	218	0	1	0	0	0	11	20	34	7	1	0	0
	PrecStd	74	3	2	2	8	16	39	44	38	24	8	6	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.7	-0.9	10.2	20.0	26.8	32.0	31.9	29.9	25.3	16.8	6.2	-4.0
		MCWB	-9.7	-4.2	3.1	9.3	14.6	19.0	20.2	19.5	14.3	8.8	1.7	-5.9
	2%	DB	-11.9	-4.5	6.6	16.0	23.6	29.0	29.5	27.0	22.1	12.7	1.7	-8.5
		MCWB	-13.0	-6.8	1.0	7.1	12.6	17.5	19.3	18.3	13.4	6.0	-1.6	-9.7
	5%	DB	-15.0	-7.8	3.3	12.8	21.1	26.8	27.6	24.9	19.4	9.8	-1.4	-12.7
		MCWB	-15.9	-9.4	-1.0	5.2	11.2	16.8	18.9	17.5	12.1	4.2	-3.7	-13.6
	10%	DB	-17.8	-10.5	0.4	10.2	18.4	24.7	25.7	22.9	16.9	7.2	-4.3	-16.1
		MCWB	-18.5	-11.8	-3.0	3.7	9.8	16.0	18.3	16.8	10.9	2.5	-6.0	-16.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-9.6	-4.0	3.6	9.7	15.1	21.2	22.4	21.0	16.7	9.1	1.6	-6.0
		MCDB	-7.8	-1.2	9.8	18.7	24.9	28.9	28.4	25.9	21.7	16.3	6.1	-4.3
	2%	WB	-13.1	-6.8	1.2	7.4	13.2	19.1	20.8	19.5	14.9	6.4	-1.3	-9.8
		MCDB	-11.9	-4.6	6.4	15.2	22.2	26.0	26.8	24.5	19.1	12.2	1.5	-8.4
	5%	WB	-15.8	-9.3	-0.9	5.5	11.7	17.9	19.8	18.5	13.4	4.5	-3.6	-13.5
		MCDB	-15.0	-7.8	3.2	12.3	19.7	24.1	25.5	23.2	18.1	9.3	-1.4	-12.7
	10%	WB	-18.4	-11.8	-2.9	3.9	10.4	16.7	18.9	17.5	11.6	2.8	-6.0	-16.6
		MCDB	-17.8	-10.6	0.3	9.9	17.7	23.1	24.1	21.8	16.4	6.7	-4.3	-16.1
Mean Daily Temperature Range	5% DB	MDBR	14.3	19.2	18.4	15.3	16.1	16.2	14.4	13.7	15.3	14.4	14.4	13.0
		MCDBR	16.1	20.1	18.7	21.0	22.4	21.2	19.0	18.0	20.6	18.5	16.1	14.8
		MCWBR	14.9	17.9	13.5	12.8	12.0	10.6	9.6	9.6	12.0	12.2	12.8	13.5
	5% WB	MCDBR	15.9	19.7	18.3	20.1	20.3	17.7	15.7	14.7	17.2	17.5	15.8	14.8
		MCWBR	14.8	17.8	13.4	12.6	11.4	9.5	8.6	8.3	10.3	11.8	12.7	13.5
Clear-Sky Solar Irradiance	taub		0.248	0.276	0.315	0.412	0.517	0.479	0.429	0.390	0.340	0.300	0.271	0.251
	taud		2.232	2.201	2.191	2.052	1.816	2.038	2.248	2.344	2.427	2.442	2.304	2.185
	Ebn at Noon		755	834	868	814	739	779	818	833	840	808	716	665
	Edh at Noon		63	89	112	148	198	160	128	110	90	71	59	55
All-Sky Solar Radiation	RadAvg		1.36	2.51	4.20	5.23	5.64	5.98	5.54	4.74	3.70	2.51	1.44	1.04
	RadStd		0.04	0.05	0.10	0.32	0.37	0.45	0.49	0.28	0.30	0.16	0.07	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page